

Public Economics and Development

Week 14: Monitoring and Incentivizing Public Sector Workers

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

May 26th, 2026

Why politicians face little scrutiny – and why it matters

Citizens hold little information on political activities and limited punishment mechanisms

- Limited channels (local radio, low-circulation press) to learn about politician behavior
 - ▶ Media and internet are often censored
- Actual data on whether and how services reach endpoints is limited
- Recall elections and audits can be misused by political rivals and pressure groups

Why accountability still matters

- Need to prevent diversion of public resources away from intended recipients
- Electoral outcomes serve as strong incentives on elected officials
 - ▶ Whereever legally feasible, politicians care about re-election
 - ▶ Even if not, continuation of incumbent party rule still matters
- Long run: Shape who enters politics, not just disciplining incumbents

Media as an accountability tool – promise and limits

How media promotes accountability

- Reward well-behaving politicians, not only punishes corrupt ones
- Disseminate information via dominant channels (radio, television, web)
- Effective bottom-up enforcement: Campaign for or against various causes
- Lower coordination costs and facilitate collective action (esp, social media and mobile)

But media can backfire very badly

- Above positive assumes sufficient media reach (not always the case)
- Misinformation can entrench incumbents or manufacture political crises
- In low-trust environments, media may not shift voter behavior at all

⇒ This is a large field in political economy that deserves its own semester

Incentivizing bureaucrats – types of instruments

Financial incentives

- Nonlinear pay tied to performance (better service delivery and showing up)
- Subject to fiscal constraints and politically difficult for entrenched civil servants

Nonmonetary measures: Job postings and career progression

- Assignment favoring high-performers can induce effort
 - ▶ Potential mismatch: preference of top performance and where they are needed may differ
- Bureaucrats care about career progression (how fast they are promoted)
 - ▶ Often linked with post-retirement benefits and social prestige
 - ▶ Inflexible trajectory in general: Could lead to them exiting for outside options
 - ▶ So they care about not getting suspensions - may lead to passive behaviors instead

Unlike private organizations, role of financial incentives could be limited

- Payscale is generally inflexible
- Side effects at recruitment: Crowd out prosocial individuals (Deserrano 2019)

Roadmap for today

1. Overview setup of the topic

2. Monitoring and Accountability

- 2.1 Reinikka, Svensson (2005): Fighting Corruption to Improve Schooling: Evidence from a Newspaper Campaign in Uganda
- 2.2 Ferraz, Finan (2008): Exposing Corrupt Politicians: The Effects of Brazil's Publicly Released Audits on Electoral Outcomes
- 2.3 Artiles, Klein-Rueschkam, León-Cilotta (2021): Accountability, Political Capture, and Selection into Politics: Evidence from Peruvian Municipalities

3. Incentives on bureaucrats

- 3.1 Duflo, Hanna, Ryan (2012): Incentives Work: Getting Teachers to Come to School
- 3.2 Khan, Khwaja, Olken (2019): Making Moves Matter: Experimental Evidence on Incentivizing Bureaucrats through Performance-Based Postings
- 3.3 Bertrand, Burgess, Chawla, Xu (2020): The Glittering Prizes: Career Incentives and Bureaucratic Performance

Reinikka, Svensson (2005):
Fighting Corruption to Improve Schooling:
Evidence from a Newspaper Campaign in
Uganda

Q: How can monitoring improve public goods and services?

How do you improve provision of public goods

- Many progress in reduced-form natural and randomized experiments
- When scaling the findings up, understanding of the entire delivery chain matters
 - ▶ What institutional constraints hinder the policy decisions?
 - ▶ What incentives influence different layers of governments, agents, and beneficiaries?
- These are required for more fruitful policy implications

This paper highlights the significance of governance structure in public services

- Using a policy experiment from newspaper campaign in Uganda
- The aim is to see how providing information reduces capture of public funds
- Induce bottom-up enforcement by publishing data on educational grant usage
- Improved primary school enrollment and performance

Educational grants and newspaper experiment in Uganda

Educational grant program aimed to support financing nonwage expenditures of schools

- Funds disbursed by Ministry of Education to the schools on regular basis
- On paper, it seemed to work well (central government budget data)
- On the ground, only 20% of the funds reached schools, many schools received nothing
- Bulk of the grants were captured by local government officials

In response, newspaper campaign was rolled out to combat these issues in late 1990s

- Not involving enhancement of legal and financial institutions (top-down)
- Campaigns induce citizen enforcement through increased awareness
- Geographical variation induced by the placement of newspaper outlet
- Knowledge of info instrumented with above geographical variation

Improvements since the newspaper campaign

TABLE 1. Summary information on capture: Grants received as share of entitled grants (in percent).

	Mean	Median	Standard deviation	Observation
All schools				
1995	23.9	0	35.1	229
2001	81.8	82.3	24.6	217

Data and empirical strategy

Sources of financial data

- Public Expenditure Tracking Survey (PETS)
 - ▶ Measures how much public resources were filtered down to *schools*
 - ▶ Central government data only shows how much *outflow* was under educational grant
- Also controls for time and income of the local region

Two ways of exploring effects of differential information

- IV: Share of funding reaching school instrumented with distance to newspaper outlet

$$\Delta \text{students}_j = \alpha + \gamma_0 \Delta x_j + \gamma_1 \sigma_{2001} + \gamma_2 \Delta \hat{s}_j + e_j$$

- DID: Variation in newspaper circulation per district across time

$$\text{students}_j = \alpha + \gamma x_j + \sigma_{1995} + \sigma_{2001} + \beta_0 \text{Newspaper}_j + \beta_1 \text{Newspaper}_j \times D_{1995} + e_j$$

Positive effects of being informed on enrollment

TABLE 3. Impact of the newspaper campaign on school enrollment.

Specification	1	2 IV regression 3		4	5
Dependent variable	Δ students	Δ s	Δ students	Δ students	Δ students
Period	95-01	95-01	95-01	95-01	91-95
Constant	450*** (20.3)	74.1*** (6.77)	29.7 (286)	574*** (49.3)	68.4** (33.6)
Distance to nearest newspaper outlet		-5.74** (2.45)		-37.7** (17.9)	-4.7 (12.1)
Share of funding reaching school			7.55* (4.62)		
Schools	202	188	188	202	153

Evidence also consistent on DID regression

TABLE 4. Difference-in-differences estimates of the effects of newspaper circulation on enrollment of grade-7 students.

Specification	1	2	3
Dependent variable	Students	Students	Students
Period	1995	2001	95, 01
1995			3,242 (328)
2001			6,988 (685)
Newspapers per school	14.5 (11.5)	34.0* (18.3)	14.0 (12.0)
Newspapers per school* 2001			20.2*** (3.18)
Schools	53	53	106

Key messages and developments

Key message: information can improve public service provision

- Bottom-up approach meant to improve monitoring
- Suggests that information can lead to improvements via accountability
- Very old paper: More recent papers have found more rigorous ways to test this idea

Developments since this paper

- Randomized audits: Reduce corruption of politicians
 - ▶ Revelation of audit results affect politicians reelection chances and image
 - ▶ See Ferraz, Finan (2008), Larreguy, Marshall, and Snyder Jr (2021)
- Collective actions: Information as mediator of protests
 - ▶ Usage of online surveys, distribution of mobile data etc
 - ▶ See Manacorda, Tesei (2020), Enikolopov, Marakin, Petrova (2020), Bursztyin et al (2021)

Ferraz, Finan (2008):
Exposing Corrupt Politicians: The Effects of
Brazil's Publicly Released Audits on Electoral
Outcomes

Q: Do disclosing information on corruption affect accountability?

Information on politicians' performance is crucial for holding them accountable

- Open information induces alignment of the interests of politicians and the public
- However, such information is not randomly assigned
- Rather, information availability is endogenous to voter efforts and community traits
- Furthermore, such information could be manipulated in absence of independent media

This paper studies effects of disclosing corruption practices on electoral outcomes

- Using anticorruption program in Brazil initiated in April 2003
- Randomly select municipalities to be audited for use of federal funds
- Audit results were disseminated publicly, including to the general public
- Compare electoral outcomes of municipalities audited pre-election vs post-election

Anticorruption program in Brazil

Rollout of anticorruption program in 2003

- Aimed at discouraging misuse of public funds and promote bottom-up monitoring
- Municipalities subject to audits determined by lottery
- If chosen, all info on federal funds transferred to the auditing body
- Then, representative auditors sent to municipal governments
 - ▶ Also receive complaints from community members and municipal councilmen
- Detailed reports on irregularities submitted to the central government
- Summary of findings posted on the web and disclosed to media sources
- Most voters receives this information via local radio stations
 - ▶ Newspaper subscription is low (42 copies per 1000)

Data and empirical strategy

Data sources are primarily audit reports released to public

- From 373 municipalities (municipalities ran by 1st-term mayors)
- Total federal funds, audited amounts, itemized list of irregularities
- Corruption: Frauds in procurements, fake receipts, embezzlement, over-invoicing
- Combined with geographical surveys and electoral outcomes of '04 local elections

Estimation leverages variation in audit timing(A_{ms}), controlling for # of irregularities(C_{ms})

$$E_{ms} = \alpha + \beta_0 C_{ms} + \beta_1 A_{ms} + \beta_2 (A_{ms} \times C_{ms}) + \gamma X_{ms} + \nu_s + e_{ms}$$

- Compare municipalities audited before ($A_{ms}=1$) to those audited after
- Control for the degree of seriousness in corruption
- Also uses variation by media source in some specifications

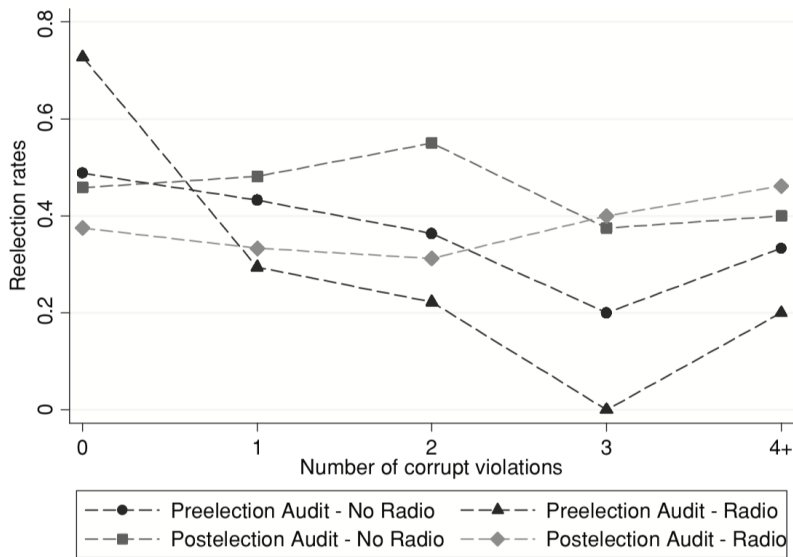
Reduction in re-election probabilities following pre-election audits

	Linear		Quadratic	Semiparametric	Corruption ≤ 5	Corruption ≤ 4
	(1)	(2)	(3)	(4)	(5)	(6)
Preelection audit	0.029 [0.083]	0.030 [0.082]	0.126 [0.101]	0.084 [0.104]	0.068 [0.087]	0.086 [0.088]
Preelection audit $\times$ number of corrupt violations	-0.038 [0.035]	-0.038 [0.035]	-0.200 [0.090]*		-0.070 [0.041]+	-0.088 [0.043]*
Preelection audit $\times$ number of corrupt violations ²			0.034 [0.017]*			
Preelection audit $\times$ corruption = 0				0.010 [0.156]		0.003 [0.036]
Preelection audit $\times$ corruption = 2				-0.253 [0.148]+		
Preelection audit $\times$ corruption = 3				-0.321 [0.192]+		
Preelection audit $\times$ corruption = 4+				-0.159 [0.168]		
Number of corrupt violations	-0.013 [0.026]	-0.012 [0.027]	0.037 [0.066]		0.012 [0.033]	0.003 [0.036]
Number of corrupt violations ²			-0.009 [0.011]			

Reduction in re-election probabilities following pre-election audits

Dependent variable: Pr(reelection)	Full sample (1)	Corruption ≤ 5 (2)	Demographic interactions (3)	Demographic and institutional interactions (4)	Households w/ radio (5)
Preelection audit	-0.059 [0.091]	-0.033 [0.096]	0.296 [1.121]	0.208 [1.247]	-0.954 [0.629]
Number of corrupt violations	-0.034 [0.029]	-0.013 [0.035]	-0.13 [0.224]	-0.069 [0.288]	-0.161 [0.194]
Number of radio stations	-0.131 [0.064]*	-0.150 [0.063]*	-0.216 [0.073]**	-0.253 [0.083]**	
Preelection audit $\times$ number of radio stations	0.229 [0.099]*	0.271 [0.104]**	0.356 [0.115]**	0.449 [0.129]**	
Preelection audit $\times$ number of corrupt violations	0.007 [0.038]	-0.018 [0.044]	-0.236 [0.402]	-0.412 [0.430]	0.458 [0.229]*
Number of corrupt violations $\times$ number of radio stations	0.050 [0.026]+	0.058 [0.025]*	0.082 [0.025]**	0.09 [0.028]**	
Preelection audit $\times$ corrupt violations $\times$ radio stations	-0.118 [0.045]**	-0.157 [0.067]*	-0.185 [0.051]**	-0.238 [0.064]**	

Results evident in simple relationships



Key findings and remaining questions

Effect of information on electoral outcomes

- Providing information on corruptions enhance accountability of politicians
- Particularly more evident if information is spread via dominant channel (radio)
- It also rewards well-behaving politicians

Remaining questions

- This is a study based on one election
- Can this effect be sustained over time?
- What about in the environment where the mass media is not trusted?
 - ▶ Or worse: When media spread misinformation?
- Political accountability in new, social media environment is actively being studied

Artiles, Klein-Rueschkam, León-Cilotta (2021):
Accountability, Political Capture, and Selection
into Politics: Evidence from Peruvian
Municipalities

Q: How does politically captured accountability mechanism function?

Can measures meant to increase accountability work when they could be manipulated?

- In principle, many mechanisms are in place to monitor politicians action and selection
 - ▶ Re-election incentives, media, recall mechanism, impeachment
- In low capacity countries, these measures could be subject to political capture
- Could be driven by special interest groups or manipulated by political elites
- Can the objectives of the accountability measure be distorted?

This paper investigates how misuse of these mechanisms affects selection of politicians

- Recall elections initiated by special interest groups in Peru
- Municipalities whose mayor barely survived recall vs ousted by small margin
- Intrinsically motivated/high-ability politicians discouraged from running
 - ▶ Recall of mayors negatively affects net value of the position

Peruvian recall procedures and political landscape

Local politics in Peru at municipal level

- Highly decentralized and responsible basic public goods (streets, security, waste)
- Mayors on 4-year terms with possibility of re-election (not after 2015)
- Very fragmented: Many independent parties based on candidates, not ideology

How recall elections work in Peru (Pretty common in Peru)

- Voters can recall local elected officials at 2nd or 3rd year of the term
- Buy recall kit (forms) → name politician → collect signature (25% of voters)
- Recalled if turn out is at least 50% and if 50% + 1 of valid votes call for recall
- No makeup election on the spot (first councilor takes over)
- 20,000 recall attempts in '97-'13, with 5,000 officials facing referendum
 - ▶ Often losers of previous elections gathering support to oust incumbent
 - ▶ Mainly in small cities (→ policymakers in Lima were unaware)
 - ▶ Also in places where where mayor won by small share (many win with 35-ish%)

Data and empirical strategy

Data sources

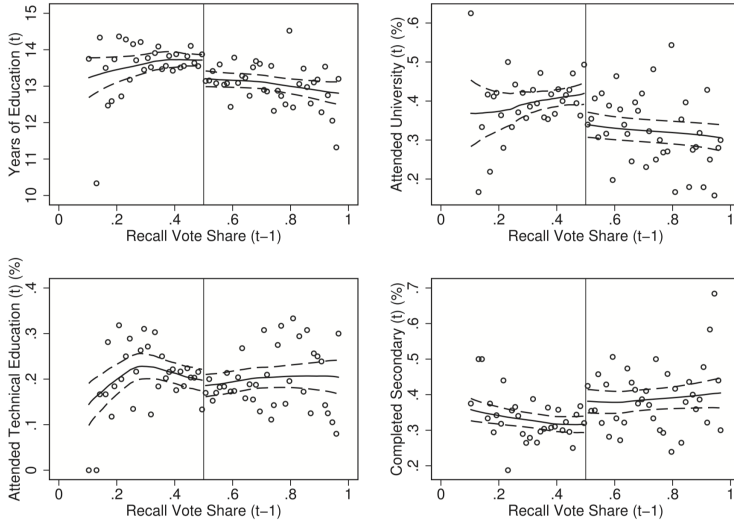
- Candidate characteristics obtained from www.infogob.com.pe
 - ▶ Maximal educational level, years of schooling, previous experience, demographics
- Electoral outcomes from national election office (ONPE)
- Budget outcomes from Ministry of Economy and Finance (MEF)

Empirical design leverages differential returns to mayorship across individuals

- Recall lowers expected term length regardless of the candidates potential performance
- Drive out policy-driven candidates and those with better outside options
 - ▶ High-ability individuals, those with policy experience ↓ Less-able ones ↑
- Leverage sharp regression design based on close recall election outcomes by vote shares

$$Y_{imt} = \alpha + \beta \text{Recalled}_{mt-1} + \gamma f(\text{voteshare}_{mt-1}) + e_{imt}$$

Decline in highly-educated candidates following referendum



Recalls are politically driven & have effects regardless of performance

	Dependent Variable			
	Years Education	University	Technical	Secondary
Panel A: Political Opponents				
Recalled Incumbent in $t - 1$	-0.2346 (0.2526)	-0.0360 (0.0347)	-0.0163 (0.0312)	0.0473 (0.0423)
Recalled in $t - 1 \times$ Political Opp. in $t - 1$	-0.5696* (0.3088)	-0.0793* (0.0436)	-0.0229 (0.0349)	0.0704 (0.0449)
Linear Polynomial	Yes	Yes	Yes	Yes
Observations	3,385	3,693	2,957	3,389
Number Municipality $\times$ Election	610	678	537	611
Mean Dep.	13,516	388	191	341
Panel B: Performance prior Recall				
Recalled Incumbent in $t - 1$	-0.5860** (0.2748)	-0.0875** (0.0379)	-0.0339 (0.0305)	0.0942** (0.0458)
Recalled in $t - 1 \times$ % Budget Executed	0.1343 (0.1337)	0.0276 (0.0250)	-0.0203 (0.0161)	-0.0068 (0.0268)
Linear Polynomial	Yes	Yes	Yes	Yes
Observations	2,565	2,791	2,237	2,565
Number Municipality $\times$ Election	384	422	336	384
Mean Dep.	13.439	0.381	0.180	0.355

Lower quality of mayors and less revenue & expenditures

	Panel A: Dependent Variable			
	Years Education	University	Technical	Secondary
Recalled Incumbent in $t - 1$	-0.1557 (0.4469)	-0.0301 (0.0861)	-0.0728 (0.0712)	0.1422 (0.0981)
Triangle Kernel	Yes	Yes	Yes	Yes
Observations	706	593	597	479
Number Municipalities	547	478	480	398
Mean Dep.	13.833	0.417	0.204	0.317

	Panel B: Dependent Variable	
	Ln(Expenditures)	Ln(Revenues)
Recalled Incumbent in $t - 1$	-0.4019** (0.1821)	-0.4902** (0.2062)
Triangle Kernel	Yes	Yes
Observations	491	421
Number Municipalities	402	356
Mean Dep.	15.964	16.259

Key findings and remaining questions

Key findings

- Accountability measures not only affect incumbents, but also potential candidates
- If the procedures are captured, there could be negative effects
 - ▶ High ability individuals discouraged from seeking positions
 - ▶ Less representative to indigenous population
 - ▶ More inefficient public finances

Remaining questions

- Mechanisms designed to give citizens control over politics could be distorted
- Particularly in cases where manipulation due to political motives are unchecked
- How to design safeguards to prevent political capture of these procedures?

Duflo, Hanna, Ryan (2012):
Incentives Work: Getting Teachers to Come to
School

Q: How does monitoring and incentives reduce absentivism?

Expansion of school access often not accompanied by improvements in school quality

- One major factor is teachers not showing up to work
 - ▶ India: about 24 % of teachers were absent during school hours
 - ▶ Para-teachers on temporary contracts often fill in for teachers' absence
- Regular teachers are politically entrenched and can resist attempts to enforce rules
- Thus, monitoring and financial incentives may work better on para-teachers

This paper tests whether monitoring and incentives reduce absenteeism of para-teachers

- Theoretically not straightforward
 - ▶ More effort for high pay vs crowd out intrinsic effort or only work until target income
- Also investigates whether schooling outcomes improve as a result
- Randomized experiment with video device and nonlinear salaries in rural India
- Provides models that shows how para-teachers respond to financial incentives

Experimental setup

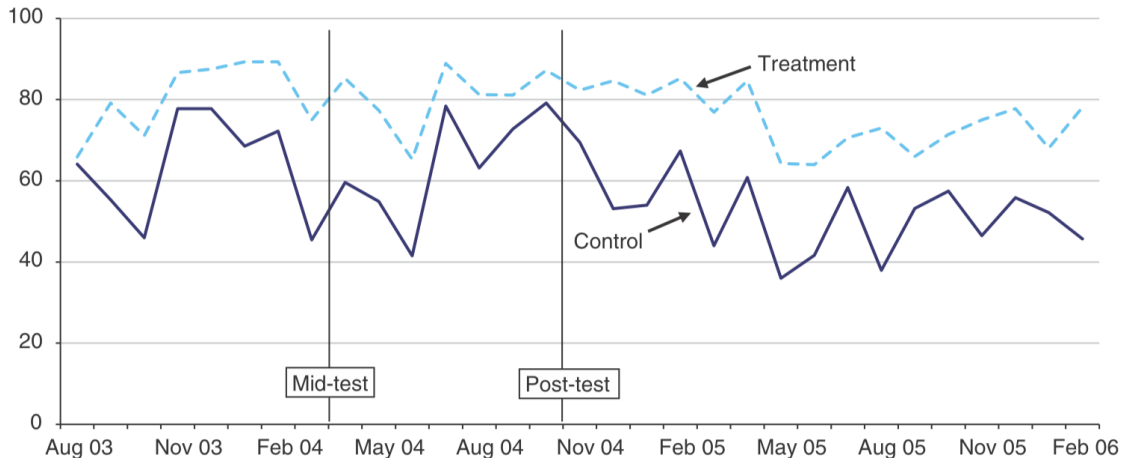
Experimental design (jointly at Udaipur organized with Seva Mandir (NGO))

- Education carried out in nonformal education centers (NFEs)
 - ▶ 1 para-teacher per NFE, teaching about 20 students
 - ▶ When the para-teacher is absent, the school is closed
- Base salary: \$ 160 PPP for at least 20 days of work per month
- in '03, the NGO assigned 60 schools in the treatment group, 60 in control
 - ▶ Teachers given camera and students instructed to take picture of teachers and students
 - ▶ Each camera came with tamper-proof settings
 - ▶ \$1.15 PPP bonus (penalty) for each additional (shortfall) day in excess of 20 days

Data collection

- Attendance: Unannounced visits and camera data
- School activities: # of children, writing on blackboard, teacher talk w/ students?
- Performance measured with 3 competency exams (pre/mid/post tests)

Teacher attendance markedly increases



Particularly in schools with teachers below median test scores

September 2003–February 2006			Difference between treatment and control schools		
Treatment (1)	Control (2)	Diff (3)	Until mid-test (4)	Mid- to post-test (5)	After post-test (6)
<i>Panel A. All teachers</i>					
0.79	0.58	0.21 (0.03)	0.20 (0.04)	0.17 (0.04)	0.23 (0.04)
1,575	1,496	3,071	882	660	1,529
<i>Panel B. Teachers with above median test scores</i>					
0.78	0.63	0.15 (0.04)	0.15 (0.05)	0.15 (0.05)	0.14 (0.06)
843	702	1,545	423	327	795
<i>Panel C. Teachers with below median test scores</i>					
0.78	0.53	0.24 (0.04)	0.21 (0.05)	0.14 (0.06)	0.32 (0.06)
625	757	1,382	412	300	670

Financial incentives do not crowd out effort

	September 2003–February 2006			Difference between treatment and control schools		
	Treatment (1)	Control (2)	Diff. (3)	Until mid-test (4)	Mid- to post-test (5)	After post-test (6)
Percent of children sitting within classroom	0.72	0.73	−0.01 (0.01)	0.01 (0.89)	0.04 (0.03)	−0.01 (0.02)
	1,239	867	2,106	643	408	983
Percent of teachers interacting with students	0.55	0.57	−0.02 (0.02)	−0.02 (0.04)	0.05 (0.05)	−0.04 (0.03)
	1,239	867	2,106	643	480	983
Blackboards utilized	0.92	0.93	−0.01 (0.01)	−0.03 (0.02)	0.01 (0.02)	−0.01 (0.02)
	990	708	1,698	613	472	613

Students's educational performance also improved

Mid-test				Post-test			
Took written (1)	Math (2)	Lang. (3)	Total (4)	Took written (5)	Math (6)	Lang. (7)	Total (8)
<i>Panel A. All children</i>							
0.04	0.15	0.16	0.17	0.06	0.21	0.16	0.17
(0.03)	(0.07)	(0.06)	(0.06)	(0.04)	(0.12)	(0.08)	(0.09)
1,893	1,893	1,893	1,893	1,760	1,760	1,760	1,760
<i>Panel B. With controls</i>							
0.04	0.13	0.14	0.14	0.06	0.18	0.14	0.15
(0.03)	(0.07)	(0.06)	(0.06)	(0.04)	0.13	0.08	0.09
1,752	1,752	1,752	1,752	1,760	1,760	1,760	1,760

Key messages and the remaining questions

Monitoring combined with financial incentives work

- Decrease teacher absenteeism and improve student outcomes
- Efforts of teachers are not crowded out while in school
- Monitoring alone not as effective as monitoring combined with financial incentives

Remaining questions

- Can this be extended to regular teachers in government schools?
 - ▶ Politically powerful and likely to resist regulations
 - ▶ How do you surmount entrenched interest to institute changes?
- Can this work in other professions and contexts?
 - ▶ Similar pilot program on local nurses not effective long-run (∴ many exemptions)
 - ▶ Tailoring monitoring/financial incentives accordingly?

Khan, Khwaja, Olken (2019):
Making Moves Matter: Experimental Evidence
on Incentivizing Bureaucrats through
Performance-Based Postings

Q: Can incentives through posting improve organizational efficiency?

Governments often aim to provide incentives through nonmonetary means

- Financial incentives are subject to regulation and fiscal constraint
- Spots for promotion are often limited, especially in seniority-based systems
- Where people are posted can function as incentives
 - ▶ Bureaucrats have preferences over different work locations and move frequently
 - ▶ In practice, posting is determined by factors other than performance
- Challenges: Diverse preferences over postings, and may not reveal truthfully

This paper asks whether assignment of bureaucrats can work as an incentive mechanism

- Large-scale experiment in property tax department in Punjab, Pakistan
- Posting purely based on performance (high-performer pick first) vs status quo
- Tests how new assignment mechanism affects tax revenues
- Identify who are most likely have stronger incentives

Assignment mechanism: Performance-ranked serial dictatorship

How it works in design

- Serial dictatorships: Individuals take turns choosing among remaining options
- Known to be able to uncover true preferences over fixed set of slots
- Early theories are silent on how the individuals should be ordered

How it is implemented here and what the theory predicts

- Ordering by performance (recovery, tax assessments)
- Increased performance $\rightarrow$ Likelihood of getting preferable slot $\uparrow$
- Incentive effects depend on own vs. others' preference, and performance distribution
 - ▶ If indiv i prefers a popular slot j - more incentives (and vice versa)
 - ▶ If i is very likely or unlikely to outperform, then lower incentives

Data and experimental setup

Background information on Punjab, Pakistan (110 million population)

- Low tax collection: low tax base, inadequate enforcement, corruption
- Collection unit (circle): led by 1 inspector and covers 2,000-10,000 properties
 - ▶ They determine how property tax formula is applied and if the property is liable
 - ▶ Size, amenities, and ease of collecting taxes vary - inspectors care about postings
- 1/3 of inspectors transferred per year: Status quo mechanism is opaque

Data sources and the 2-year experiment

- Tax revenues, base, recovery rates, non-exemption rates collected at circle level
- Preference data from all inspectors in all circles (rank all circles within a group)
 - ▶ Collected before the announcement of treatment assignment
- Randomization at group level (10 circles) for each year by a lottery
 - ▶ Performance based on recovery (1/2 of treatment) and tax assessments (other half)

Improved tax revenue following experiment

	Year 1	Year 2	Pooled
	(1)	(2)	(3)
Treatment	0.049 (0.022) [0.009]	0.092 (0.042) [0.036]	0.061 (0.020) [0.002]
Baseline	0.892 (0.018)	0.946 (0.019)	0.944 (0.019)
Observations	405	251	656
Mean growth in controls	0.117	0.309	0.203

But the effect dies off with the repeated treatment

	Total (1)	Current (2)	Arrears (3)
Y1 treatment (β_1)	0.109 (0.038) [0.003]	0.085 (0.040) [0.020]	0.128 (0.100) [0.200]
Y2 treatment (β_2)	0.081 (0.043) [0.064]	0.055 (0.041) [0.235]	-0.074 (0.119) [0.592]
Y1 and Y2 treatment (β_3)	-0.150 (0.067) [0.014]	-0.085 (0.068) [0.203]	-0.061 (0.178) [0.733]
Observations	403	403	392
$\beta_1 = \beta_2$	0.564	0.560	0.167
$\beta_1 + \beta_3 = 0$	0.401	0.999	0.655

Treatment effect varies with knowledge of preference and outcomes

Estimated Treatment $\times$ model-predicted effort	Revenue	
	(1)	(2)
Full knowledge of $\mathbf{P}$, $\mathbf{y}$ (know others' pref & performance distribution)	0.038 (0.051)	0.042 (0.066)
Random $\mathbf{P}$, full knowledge of $\mathbf{y}$ (know performance distribution, not others' pref)	0.076 (0.036)	0.176 (0.103)
Assume identical $\mathbf{P}$, full knowledge of $\mathbf{y}$ (assume same pref as own, know performance distribution)	0.025 (0.027)	0.051 (0.051)
Full knowledge of $\mathbf{P}$, no knowledge of $\mathbf{y}$ (know others' pref, not performance distribution)	-0.009 (0.060)	-0.014 (0.069)
Observations	249	249
Mean of control group	16.268	16.268

Key findings and remaining questions

Other key findings

- The effect sizes are similar to the performance pay experiment
 - ▶ Carried out in similar contexts by the same group of authors!
- Group that increases the effort most?
 - ▶ Case where others' performance are predictable but not preferences
 - ▶ Uncertainty about preferences + info on how much to improve → effort to raise rank ↑

Remaining questions

- Nonpecuniary incentives can work as much as pecuniary ones
- Both schemes may reduce flexibility over assignment
 - ▶ Sometimes, may need to send top performers where they are needed most
 - ▶ This may be different from agents' preferences
- How to balance organizational goals with individual motivation?

Bertrand, Burgess, Chawla, Xu (2020):
The Glittering Prizes: Career Incentives and
Bureaucratic Performance

Q: How does career incentives of bureaucrats affect performance?

What motivates bureaucrats to work hard?

- We know how politicians and private sector managers affect performance
- But what motivates bureaucrats?
 - ▶ They enter through a competitive procedures (exams)
 - ▶ They are seldom fired, promoted based on seniority, and retire at fixed age
- Meant to limit favoritism and prevent lobbying, but where does motivation kick in?

This paper investigates this question using survey and administrative data on bureaucrats

- Using examples from Indian Administrative Service (IAS)
- What motivates IAS workers? Promotion to top positions and perks that come with it
- Leverage natural experiment from variation in age at entry
 - ▶ Those who entered old cannot reach the top → Are they less motivated?
 - ▶ Also leverage pension reform that extended retirement age by 2 years

Who are the IAS officers?

Elite administrative civil service of India (that shaped other Commonwealth countries)

- Competitive entry based on exams: 120 acceptances for roughly 450,000 applicants
- Age limit: Must be 21-30 years old at entry
- Selected officers allocated to state cadre in the subsequent year based on rules

What are their generic career progressions?

- Promotion by seniority: After 4, 9, 13, 16, 25 and 30 years to reach next payscale
 - ▶ Salary jump at final payscale: 25-60%
- Actual tenure required for promotion is stringent for later promotions
 - ▶ Depends on performance reviews, and availability
- Retirement at 58 before '98 and 60 post-'98, with pensions linked to final payscale
- Retiring at highest payscale comes with social prestige and other nonpecuniary perks
- Those who enter older may not reach top payscale → Incentives and effectiveness ↓

Data and regression analysis

Measuring bureaucratic performance

- Surveys assessing bureaucrats from 14 states (85% of Indian pop.) on 1-to-5 scale
 - ▶ Overall rating, job effectiveness, probity, withstand illegal pressure, pro-poor sentiment
 - ▶ From other IAS officers, state civil servants, politicians, businessmen, media, civil society
 - ▶ Subjectivity addressed with source FE, interviewer FE, and respondent FE
- Names of IAS workers compiled at state level - covers 70% of all officers
- Administrative data: Gather suspension episodes, internal rankings etc

Regression leverage variation in age of entry and timing of reforms at '98

- Leveraging entry age only (agent i , respondent j):

$$y_{ij} = \alpha \text{age_entry}_i + \beta X_i + \theta_j + e_{ij}$$

- Leveraging entry age and reforms

$$y_{ij} = \alpha_1 \text{age_entry}_i + \alpha_2 \text{age_entry}_i \times \text{post98}_i + \beta X_i + \theta_j + e_{ij}$$

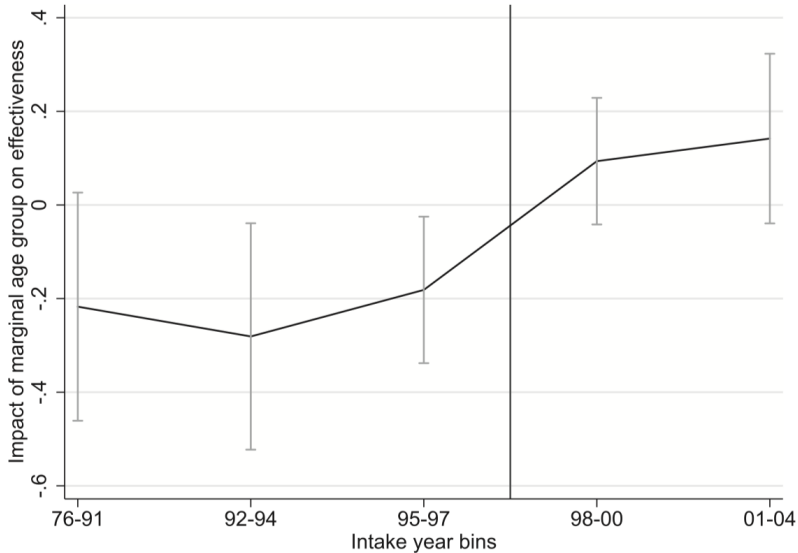
Those who enter older are less likely to be perceived as effective

Panel A	(1)	(2)	(3)	(4)	(5)
			Effectiveness (1 low, 5 high)		
Mean of dep. var	3.734	3.734	3.734	3.734	
Age at entry	-0.014*** (0.005)	-0.013** (0.005)	-0.013** (0.006)		0.019 (0.018)
Reaching top				0.118*** (0.044)	0.253* (0.138)
Respondent FEs	Y	Y	Y	Y	Y
State × Intake year FEs		Y	Y	Y	Y
Controls			Y	Y	Y
Sample			Pre-reform cohorts		
Observations	15,396	15,396	15,396	15,396	15,396
Panel B	(1)	(2)	(3)	(4)	(5)
	Effective	Probity	Pressure	Pro-poor	Overall
Mean of dep. var	3.734	3.677	3.526	3.533	3.646
Age at entry	-0.013** (0.006)	-0.008 (0.007)	-0.014** (0.005)	-0.004 (0.005)	-0.011* (0.006)
Respondent FEs	Y	Y	Y	Y	Y
State × Intake year FEs	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y
Sample			Pre-reform cohorts		
Observations	15,396	13,129	14,512	14,782	15,340

Delayed retirement increases perceived effectiveness of older entrants

Panel A	(1)	(2)	(3)	(4)	(5)
	Effectiveness (1 low, 5 high)				
Mean of dep. var	3.734	3.701	3.730	3.730	3.730
Age at entry	-0.013** (0.006)	0.027* (0.016)	-0.012** (0.006)	-0.040*** (0.013)	-0.040*** (0.012)
Age at entry × Post-reform cohort			0.031*** (0.011)	0.066*** (0.018)	0.067*** (0.020)
Respondent FEs	Y	Y	Y	Y	Y
State × Intake year FEs	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y
Age at entry × Intake year				Y	Y
Controls × Post-reform cohort					Y
Sample	Pre	Post	Pre- and post-reform cohorts		
Observations	15,396	2,186	17,749	17,749	17,749
Panel B	(1)	(2)	(3)	(4)	(5)
	Effective	Probity	Pressure	Pro-poor	Overall
Mean of dep. var	3.730	3.671	3.524	3.528	3.647
Age at entry	-0.040*** (0.012)	-0.049*** (0.016)	-0.040*** (0.013)	-0.022* (0.013)	-0.033** (0.014)
Age at entry × Post-reform cohort	0.067*** (0.020)	0.076*** (0.024)	0.039** (0.019)	0.029 (0.020)	0.057*** (0.021)
Respondent FEs	Y	Y	Y	Y	Y
State × Intake year FEs	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y
Age at entry × Intake year	Y	Y	Y	Y	Y
Controls × Post-reform cohort	Y	Y	Y	Y	Y
Sample	Pre- and post-reform cohorts				
Observations	17,749	15,133	16,717	17,042	17,694

Performance of marginal entrants (age 28,29) improves post-reform



Takeaways and remaining questions

Key findings

- Paper also finds that older entrants working with younger colleagues perform worse
- Older entrants with less likelihood of reaching top levels face more suspensions
- All evidence suggest career potentials act as incentive mechanism for bureaucrats
- Thus highlights the role of nonmonetary instruments in shaping performance

Remaining questions

- How do civil leaders affect societal outcomes?
 - ▶ Paper is silent on economic and social outcomes of motivated bureaucrats
 - ▶ Ultimate guidance on whether bureaucratic reforms work!
- Roles of other nonmonetary instruments?
 - ▶ Social prestige that comes with bureaucracy, for instance
 - ▶ Potentially leverage exogenous changes in the exams selecting bureaucrats?

Unanswered questions

Citizens as a stakeholder in all of this?

- What is the best way to monitor bureaucracy (media, SNS, civil society etc)
- What method of public organizations serves the citizens the best?
- How do you motivate them to join public organizations (or comply to public policies)?

Ultimately, what is the best way to organize a bureaucracy?

- Typically organized in hierarchy
- But alternatively, some are form based on tasks or events (e.g. task forces)
- Is clear structure better (may need bureaucrats to carry difficult tasks against will)
- Or is providing flexibility better?

Keep an interest on Vorada's class entirely dedicated to bureaucracy next semester